

Inside BDH

Interactive microscope for sparse, non-negative neural activity

DataForge 2026 - Pathway Track | Artifact: <https://inside-bdh-1.onrender.com> | Source: <https://github.com/himani27301/inside-bdh>

CENTRAL CLAIM: BDH converts a signed latent projection into an input-dependent, non-negative sparse state; changing the input changes which latent units survive.

Concept and mechanism

Sparse activation means many candidate internal units are inactive for a particular input while a smaller subset carries non-zero activity. Non-negativity means surviving values are zero or positive. These properties are related but distinct: ReLU guarantees non-negativity and zeros all non-positive pre-activations, while the amount and structure of sparsity still depend on learned weights and the input. The BDH paper reports sparse, positive activation vectors and presents state interpretability as a design goal.[1] The public toy implementation contains $\mathbf{x_latent} = \mathbf{x @ encoder}$, followed by $\mathbf{x_sparse} = \text{ReLU}(\mathbf{x_latent})$. Inside BDH instruments both tensors, so the learner sees signed pre-ReLU values, the post-ReLU state, per-layer activity, head activity, and strongest last-token latent units. The interface therefore exposes the real forward pass rather than a scripted animation

Ground-truth check
Input: "The king walked into the castle." Layer 1 has **82.05%** non-positive $\mathbf{x_latent}$ values and exactly **82.05%** zero $\mathbf{x_sparse}$ values. The same run leaves **17.95%** of Layer-1 $\mathbf{x_sparse}$ units active. The expected ReLU rule and observed zeroing match.

Method	Sparse state	Interpretability
Toy BDH	ReLU on learned latent projection.	Direct pre/post state inspection.
Sparse autoencoder	Sparse bottleneck/dictionary.[2][3]	Post-hoc feature latents.
Transformer baseline	Sparsity not defining constraint.	Attention/probes/post-hoc tools.

Input-dependent state and evidence

With the same checkpoint, Layer-1 $\mathbf{x_sparse}$ activity changes from **17.95%** for a natural sentence to **20.29%** for repetition and **42.52%** for a noise-like string. Top-20 last-token unit overlap with the natural sentence is 20% for repetition and 0% for noise; another natural sentence in our small test shares 100%. These are local measurements of this independently trained byte-level Tiny-Shakespeare checkpoint, not general BDH guarantees. **Live evidence** in the artifact includes the pre/post-ReLU tensors, activity rates, A/B comparison, and unit overlap. **Published developer evidence** from the BDH paper includes sparse positive activations and broader state-interpretability claims.[1] Related sparse-autoencoder research shows that sparse latent spaces can support feature-level interpretability, but SAE features are not the same mechanism as BDH neurons or synapses.[2][3]

BDH, BDH-CQ, maturity, and limitations

BDH is the direct substrate in this artifact. The 2025 paper describes a brain-inspired architecture with sparse positive activations, synaptic-memory mechanisms, and state interpretability.[1] BDH-CQ is a later family member that combines in-context learning with recurrent latent reasoning.[4] This demo does **not** run BDH-CQ and does not claim to reproduce its ARC-AGI results. BDH is best treated as **research-stage**: technically interesting and publicly documented, but not yet a mature drop-in replacement for mainstream Transformer systems. The strongest property of Inside BDH is that the learner can predict ReLU output, change the input, and inspect the resulting state. Its main limitation is evidence scale: the checkpoint is small, byte-level, and trained on Tiny Shakespeare. ReLU explains zeroing and non-negativity, but does not by itself establish monosemanticity, scale-free connectivity, synaptic-memory quality, or every sparsity property reported in larger BDH runs. Top-unit overlap is an educational diagnostic, not semantic proof.

Primary sources (2022-2026)

[1] Kosowski, A. et al. (2025). The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. arXiv:2509.26507.
[2] Cunningham, H. et al. (2023). Sparse Autoencoders Find Highly Interpretable Features in Language Models. arXiv:2309.08600.
[3] Gao, L. et al. (2024). Scaling and Evaluating Sparse Autoencoders. arXiv:2406.04093.
[4] Engdahl, B. et al. (2026). BDH-CQ: In-Context Learning with Recurrent Latent Reasoning. arXiv:2608.09888.